Capstone Project

Data-Driven Readmission Risk Prediction for Improved Diabetes Care Coordination

Final Report

1. **Define the Problem Statement:**

   Hospital readmissions among diabetic patients are a persistent challenge for healthcare systems, contributing to increased healthcare costs, patient complications, and financial penalties under value-based care programs. Many readmissions are potentially preventable if high-risk patients can be identified early and provided with targeted care management interventions before and after discharge. However, clinicians often lack tools that effectively synthesize patient demographic, clinical, and utilization data to estimate an individual patient's likelihood of readmission.

   The goal of this project is to develop a machine learning–based approach to **predict unplanned 30-day hospital readmission risk among diabetic patients using data available at discharge**. By leveraging historical healthcare utilization patterns and indicators of patient complexity, the model aims to identify patients who are most likely to benefit from enhanced follow-up care, medication management, and care coordination.

   The primary challenge lies in accurately capturing the complex relationships between patient characteristics, prior healthcare utilization, and clinical complexity that contribute to readmission risk. Additionally, the model must balance predictive performance with interpretability so that its insights can be trusted and used by healthcare providers.

   If successfully implemented, this solution could help healthcare organizations **prioritize high-risk patients, allocate care management resources more effectively, and reduce avoidable readmissions**, ultimately improving patient outcomes while supporting the goals of value-based healthcare delivery.

2. **Model Outcomes or Predictions:**

   This project primarily uses **classification**, as the goal is to predict whether a diabetic patient will experience an **unplanned hospital readmission within 30 days**. The target variable is binary, indicating either **readmitted (1)** or **not readmitted (0)**. The expected output of the model is therefore a **probability score representing the likelihood of readmission**, which can then be converted into a classification label indicating whether a patient is predicted to be at high or low risk of readmission.

The project primarily uses **supervised learning algorithms**, since the dataset contains labeled outcomes indicating whether each patient was readmitted. Several supervised classification models were trained and compared, including **Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, and K-Nearest Neighbors**.

In addition to supervised learning, **unsupervised learning** was also applied using **K-Means clustering** to segment patients into groups based on healthcare utilization patterns and clinical characteristics. This approach helps identify distinct patient risk segments that may benefit from targeted care management interventions.

3. **Data Acquisition:**

To develop a machine learning model for predicting **30-day hospital readmission among diabetic patients**, the analysis uses a dataset containing patient demographic, clinical, and healthcare utilization information available at discharge. The dataset includes features that capture both patient characteristics and patterns of prior healthcare use, which are known drivers of readmission risk.

Key data categories used in this project include:

Patient Demographics

- Age group

Clinical and Treatment Characteristics

- Number of medications
- Number of laboratory procedures
- Number of medical procedures
- Length of hospital stay

Healthcare Utilization History

- Number of prior inpatient visits
- Number of emergency visits
- Number of outpatient visits
- Total prior visits (engineered feature)

Target Variable

- 30-day hospital readmission (binary outcome)

These features were selected because they reflect **patient complexity, prior utilization patterns, and treatment intensity**, all of which are commonly associated with hospital readmissions.

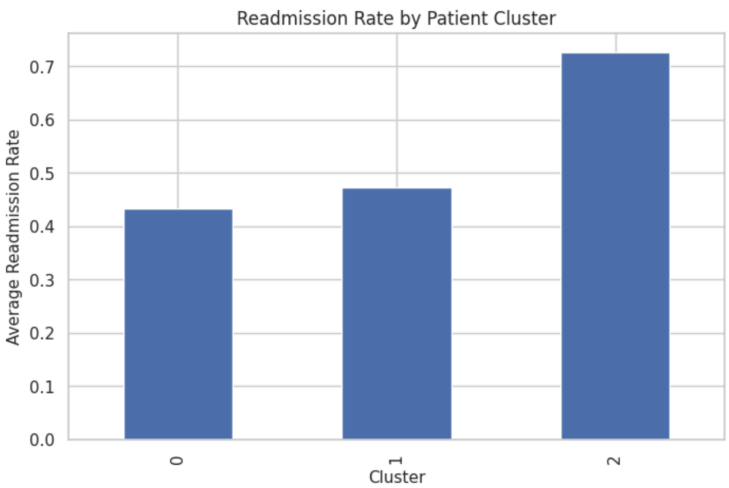
In a real-world healthcare analytics setting, additional data sources could further strengthen the model, including:

- Electronic Health Records (EHR) for detailed clinical history
- Claims data for longitudinal utilization patterns
- Social determinants of health (e.g., socioeconomic status, access to care)
- Medication adherence data
- Comorbidity indices such as the Charlson Comorbidity Index

Incorporating multiple data sources would help capture a more complete view of patient health and social risk factors that influence readmission outcomes.

Visualization to Assess Data Potential

To evaluate whether the available data could help predict readmission risk, exploratory analysis was conducted to examine the relationship between **prior healthcare utilization and readmission outcomes**. The following visualization illustrates how readmission rates vary across patient clusters identified through unsupervised learning:



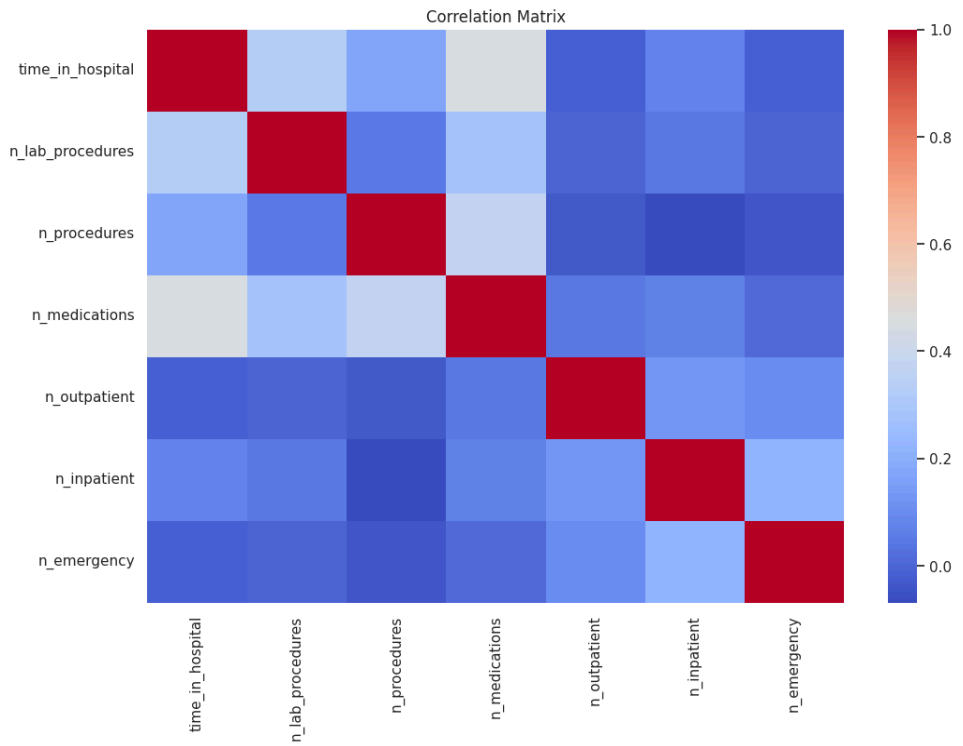
Readmission Rate by Patient Risk Cluster

- Cluster 0 represents lower-risk patients with shorter hospital stays, fewer medications, and minimal prior healthcare visits, indicating relatively lower clinical complexity.
- Cluster 1 reflects patients with longer hospital stays and higher treatment intensity, suggesting moderate clinical complexity and potential readmission risk.
- Cluster 2 shows markedly higher prior healthcare utilization and inpatient visits, indicating a group of high-utilization patients who may represent the highest-risk population for readmission and could benefit most from targeted care management interventions.

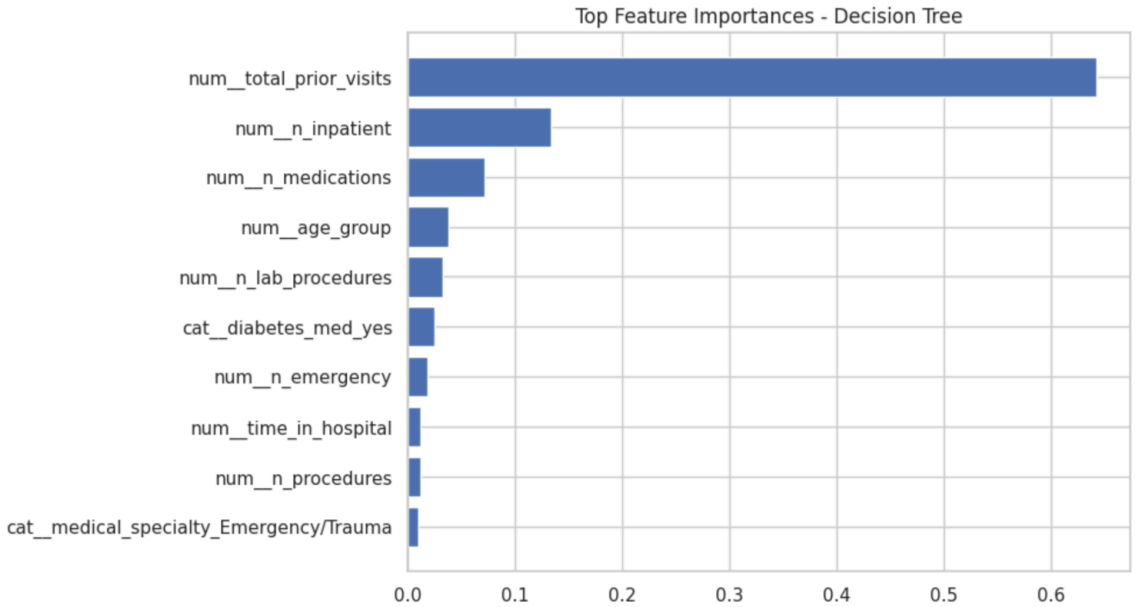
This visualization highlights how patients with higher prior healthcare utilization tend to exhibit **substantially higher readmission rates**, indicating that utilization-based features are strong predictors of readmission risk.

Here are a few additional visualizations:

Correlation heatmap of numerical variables



Feature importance rankings from predictive models



These visualizations help confirm that the dataset contains **meaningful signals that can be leveraged by machine learning models** to identify patients at risk of readmission.

4. **Data Preprocessing/Preparation:**

Before training machine learning models, several preprocessing steps were performed to ensure the dataset was clean, consistent, and suitable for modeling.

Data Cleaning and Consistency Checks

The dataset was first examined to understand its structure, data types, and potential data quality issues. Missing values were assessed using summary statistics and null value checks. The dataset did not contain significant missing values in the key modeling variables, so no imputation techniques were required. Column names were standardized and variables were reviewed to ensure consistent formats across the dataset.

Outliers and extreme values were examined during exploratory data analysis. In healthcare datasets, extreme utilization values often represent clinically meaningful cases rather than data errors, so these values were retained rather than removed.

Feature Engineering

Additional features were created to better capture patient healthcare utilization patterns. For example, a **total_prior_visits** feature was engineered by combining outpatient, emergency, and inpatient visit counts. Age groups

were also converted into an **ordinal numerical representation** to allow models to use age information more effectively.

Encoding and Scaling

Since the dataset contained both numerical and categorical variables, preprocessing steps were applied to prepare the features for machine learning algorithms:

- **Categorical variables** were encoded using **one-hot encoding** to convert them into numerical representations suitable for modeling.
- **Numerical variables** were standardized using **StandardScaler** to ensure features were on a comparable scale. This step is particularly important for algorithms such as Logistic Regression and K-Nearest Neighbors that are sensitive to feature magnitudes.

These preprocessing steps were implemented within a **scikit-learn pipeline**, ensuring that transformations were applied consistently during model training and evaluation while preventing data leakage.

Train-Test Split

To evaluate model performance on unseen data, the dataset was divided into **training and testing sets**. A **stratified train-test split** was used to preserve the distribution of the target variable (readmitted vs. not readmitted) across both sets. Approximately **80% of the data was used for training** and **20% for testing**. Stratification helps ensure that both sets maintain a similar proportion of readmission outcomes, which is important for fair model evaluation.

Summary

Through careful data cleaning, feature engineering, encoding, scaling, and stratified splitting, the dataset was prepared to support reliable model training and evaluation. These preprocessing steps helped ensure that the machine learning models could effectively learn patterns associated with hospital readmission risk while minimizing bias and data leakage.

5. **Modeling:**

To address the problem of predicting unplanned 30-day hospital readmissions among diabetic patients, several machine learning algorithms were selected and evaluated. The modeling approach began with a **baseline algorithm** and expanded to include additional models that could capture more complex relationships in the data.

- Baseline Model: Logistic Regression

Logistic Regression was chosen as the baseline model because it is widely used for binary classification problems and provides a strong level of interpretability, which is important in healthcare settings. Logistic regression estimates the probability that a patient will be readmitted based on a set of input features, allowing clinicians to understand how different factors influence risk. Regularization was applied to reduce the likelihood of overfitting and improve model stability.

- Decision Tree

  A Decision Tree classifier was evaluated as a non-linear model capable of capturing interactions between variables. Decision trees can automatically model complex decision rules and are relatively easy to interpret, making them useful for understanding how combinations of patient characteristics influence readmission risk.

- Random Forest

  Random Forest, an ensemble learning method that combines multiple decision trees, was implemented to improve predictive performance and reduce overfitting. By averaging predictions from many trees trained on different subsets of the data, Random Forest can capture more complex relationships and improve model robustness.

- Gradient Boosting

  Gradient Boosting was also evaluated as an advanced ensemble method that sequentially builds decision trees to correct errors from previous models. This approach often produces strong predictive performance on structured datasets and can capture subtle nonlinear patterns in the data.

- K-Nearest Neighbors (KNN)

  K-Nearest Neighbors was included to provide a comparison with a **distance-based classification approach**. Unlike tree-based models, KNN predicts outcomes based on similarity between observations in feature space, offering an alternative modeling perspective.

Model Evaluation

All models were trained using the training dataset and evaluated on the test dataset using **accuracy, recall, and ROC-AUC**. Particular emphasis was placed on recall to ensure the models effectively identify patients at high risk of readmission, which is especially important in a value-based healthcare context where missing high-risk patients can lead to poor outcomes and increased costs.

This combination of baseline, tree-based, ensemble, and distance-based models provided a comprehensive comparison of machine learning approaches for predicting hospital readmission risk.

6. **Model Evaluation:**

This project primarily focused on **classification models**, since the goal was to predict whether a diabetic patient would experience an **unplanned hospital readmission within 30 days** (a binary outcome: readmitted vs. not readmitted). Several supervised classification algorithms were evaluated, including **Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and K-Nearest Neighbors**. In addition to supervised learning, **unsupervised learning** using **K-Means clustering** was applied to segment patients into risk groups based on healthcare utilization patterns and clinical characteristics.

To evaluate model performance, three primary metrics were used:

- **Accuracy:** Measures the overall proportion of correct predictions made by the model.
- **Recall:** Measures the model's ability to correctly identify patients who were actually readmitted. This metric was particularly important because missing high-risk patients could lead to missed opportunities for intervention.
- **ROC-AUC:** Evaluates the model's ability to distinguish between patients who will and will not be readmitted across different classification thresholds.

The models were trained on the training dataset and evaluated on a held-out test dataset to ensure unbiased performance assessment. Hyperparameter tuning was also performed using **GridSearchCV** to optimize model parameters and improve predictive performance.

Among the models evaluated, **Gradient Boosting achieved the strongest overall performance**, with the highest ROC-AUC and competitive accuracy, indicating strong ability to distinguish between high-risk and low-risk patients. **Random Forest achieved the highest recall**, demonstrating its ability to identify the largest proportion of patients who were actually readmitted. Simpler models such as Logistic Regression and Decision Tree showed moderate performance, while K-Nearest Neighbors performed less effectively for this dataset.

Based on these results, **Gradient Boosting was selected as the most optimal model**, as it provided the best overall balance between predictive accuracy

and the ability to distinguish between readmission outcomes while maintaining robust performance on unseen data.